

ENVIRONMENTAL CHEMISTRY

ENVIRONMENTAL POLLUTION

- Environmental pollution is the effect of undesirable changes in our surroundings that have harmful effects on plants, animals and human beings.
A substance, which causes pollution, is known as **pollutant**
- Pollutants can be solid, liquid or gaseous substances present in greater concentration than in natural abundance and are produced due to human activities or due to natural happenings.
- Pollutants can be **degradable**, like discarded vegetables which rapidly break down by natural processes.
- pollutants which are **slowly degradable, remain in the environment in an unchanged form for many decades.**
- For example, substances such as **dichlorodiphenyltrichloroethane (DDT), plastic materials, heavy metals, many chemicals, nuclear wastes** etc., once released into the environment are difficult to remove.

ATMOSPHERIC POLLUTION

- ❖ The lowest region of atmosphere in which the human beings along with other organisms live is called troposphere. It extends up to the height of ~ 10 km from sea level
- ❖ Above the troposphere, between 10 and 50 km above sea level lies stratosphere.
- ❖ Troposphere is a turbulent, dusty zone containing air, much water vapour and clouds. This is the region of strong air movement and cloud formation.
- ❖ The stratosphere, contains dinitrogen, dioxygen, ozone and little water vapour
- ❖ Atmospheric pollution is generally studied as **tropospheric and stratospheric pollution**
- ❖ The presence of ozone in the stratosphere prevents about 99.5 per cent of the sun's harmful ultraviolet (UV) radiations from reaching the earth's surface and thereby protecting humans and other animals from its effect.

Tropospheric Pollution

Tropospheric pollution occurs due to the presence of undesirable solid or gaseous particles in the air.

The following are the major gaseous and particulate pollutants present in the troposphere:

- 1. Gaseous air pollutants:** These are oxides of sulphur, nitrogen and carbon, hydrogen sulphide, hydrocarbons, ozone and other oxidants.
- 2. Particulate pollutants:** These are dust, mist, fumes, smoke, smog etc.

1. Gaseous air pollutants

(a) Oxides of Sulphur:

- ✓ Oxides of sulphur are produced when sulphur containing fossil fuel is burnt.
- ✓ sulphur dioxide, is a gas that is poisonous to both animals and plants
- ✓ even a low concentration of sulphur dioxide causes respiratory diseases e.g., asthma, bronchitis, emphysema in human beings.
- ✓ Sulphur dioxide causes irritation to the eyes, resulting in tears and redness
- ✓ High concentration of SO₂ leads to stiffness of flower buds which eventually fall off from plants.
- ✓ Uncatalysed oxidation of sulphur dioxide is slow.

- ✓ However, the presence of particulate matter in polluted air catalyses the oxidation of sulphur dioxide to sulphur trioxide

(b) Oxides of Nitrogen:

- ✓ Dinitrogen and dioxygen are the main constituents of air. These gases do not react with each other at a normal temperature.
- ✓ At high altitudes when lightning strikes, they combine to form oxides of nitrogen
- ✓ NO₂ is oxidised to nitrate ion, NO₃⁻ – which is washed into soil, where it serves as a fertilizer.
- ✓ In an automobile engine, (at high temperature) when fossil fuel is burnt, dinitrogen and dioxygen combine to yield significant quantities of nitric oxide (NO) and nitrogen dioxide (NO₂)
- ✓ Rate of production of NO₂ is faster when nitric oxide reacts with ozone in the stratosphere.
- ✓ The irritant red haze in the traffic and congested places is due to oxides of nitrogen.
- ✓ Higher concentrations of NO₂ damage the leaves of plants and retard the rate of photosynthesis.
- ✓ Nitrogen dioxide is a lung irritant that can lead to an acute respiratory disease in children.
- ✓ It is toxic to living tissues also. Nitrogen dioxide is also harmful to various textile fibres and metals.

(c) Hydrocarbons:

- ✓ Hydrocarbons are composed of hydrogen and carbon only and are formed by incomplete combustion of fuel used in automobiles
- ✓ Hydrocarbons are carcinogenic, i.e., they cause cancer
- ✓ They harm plants by causing ageing, breakdown of tissues and shedding of leaves, flowers and twigs.

(d) Oxides of Carbon

(i) Carbon monoxide:

- Carbon monoxide (CO) is one of the most serious air pollutants.
- It is a colourless and odourless gas, highly poisonous to living beings because of its ability to block the delivery of oxygen to the organs and tissues
- It is produced as a result of incomplete combustion of carbon
- Carbon monoxide is mainly released into the air by automobile exhaust.
- Other sources, which produce CO, involve incomplete combustion of coal, firewood, petrol, etc.

Why carbon monoxide is poisonous?

- ❖ It binds to haemoglobin to form carboxyhaemoglobin, which is about 300 times more stable than the oxygen-haemoglobin complex.

- ❖ In blood, when the concentration of carboxyhaemoglobin reaches about 3–4 per cent, the oxygen carrying capacity of blood is greatly reduced.
- ❖ This oxygen deficiency, results into headache, weak eyesight, nervousness and cardiovascular disorder.
- ❖ This is the reason why people are advised not to smoke.
- ❖ In pregnant women who have the habit of smoking the increased CO level in blood may induce premature birth, spontaneous abortions and deformed babies.

(ii) Carbon dioxide:

- Carbon dioxide (CO₂) is released into the atmosphere by respiration, burning of fossil fuels for energy, and by decomposition of limestone during the manufacture of cement.
- It is also emitted during volcanic eruptions.
- Carbon dioxide gas is confined to troposphere only.
- Normally it forms about 0.03 per cent by volume of the atmosphere.
- With the increased use of fossil fuels, a large amount of carbon dioxide gets released into the atmosphere.
- Excess of CO₂ in the air is removed by green plants and this maintains an appropriate level of CO₂ in the atmosphere.
- Green plants require CO₂ for photosynthesis and they, in turn, emit oxygen, thus maintaining the delicate balance.
- deforestation and burning of fossil fuel increases the CO₂ level and disturb the balance in the atmosphere. *The increased amount of CO₂ in the air is mainly responsible for global warming.*

Global Warming and Greenhouse Effect

- About 75 % of the solar energy reaching the earth is absorbed by the earth's surface, which increases its temperature.
- The rest of the heat radiates back to the atmosphere.
- Some of the heat is trapped by gases such as carbon dioxide, methane, ozone, chlorofluorocarbon compounds (CFCs) and water vapour in the atmosphere.
- Thus, they add to the heating of the atmosphere. This causes global warming.
- atmosphere traps the sun's heat near the earth's surface and keeps it warm. This is called natural greenhouse effect because it maintains the temperature and makes the earth perfect for life.
- carbon dioxide molecules also trap heat as they are transparent to sunlight but not to the heat radiation.

- If the amount of carbon dioxide crosses the delicate proportion of 0.03 per cent, the natural greenhouse balance may get disturbed.
- Carbon dioxide is the major contributor to global warming
- Besides carbon dioxide, other greenhouse gases **are methane, water vapour, nitrous oxide, CFCs and ozone.**
- **Methane** is produced naturally when vegetation is burnt, digested or rotted in the absence of oxygen.
- Large amounts of methane are released in paddy fields, coal mines, from rotting garbage dumps and by fossil fuels.
- **Chlorofluorocarbons (CFCs)** are man-made industrial chemicals used in air conditioning etc.
- CFCs are also damaging the ozone layer (Section 14.2.2). Nitrous oxide occurs naturally in the environment.
- In recent years, their quantities have increased significantly due to the use of chemical fertilizers and the burning of fossil fuels
- If these trends continue, the average global temperature will increase to a level which may lead to melting of polar ice caps and flooding of low lying areas all over the earth.
- Increase in the global temperature increases the incidence of infectious diseases like dengue malaria, yellow fever, sleeping sickness, etc.

What can we do to reduce the rate of global warming?

- minimise the use of automobiles. one can use bicycle, public transport system, or go for carpool.
- plant more trees to increase the green cover.
- Avoid burning of dry leaves, wood etc. It is illegal to smoke in public places and work places

Acid rain

- When the pH of the rain water drops below 5.6, it is called acid rain
- Acid rain refers to the ways in which acid from the atmosphere is deposited on the earth's surface.
- Oxides of nitrogen and sulphur which are acidic in nature can be blown by wind along with solid particles in the atmosphere and finally settle down either on the ground as dry deposition or in water, fog and snow as wet deposition.
- Acid rain is a byproduct of a variety of human activities that emit the oxides of sulphur and nitrogen in the atmosphere
- burning of fossil fuels (which contain **sulphur and nitrogenous** matter) such as coal and oil in power stations and furnaces or petrol and diesel in motor engines produce sulphur dioxide and nitrogen oxides
- **SO₂ and NO₂** after oxidation and reaction with water **are major contributors to acid rain**

- Aerosol particles of oxides or ammonium salts in rain drops result in wet deposition.
- SO₂ is also absorbed directly on both solid and liquid ground surfaces and is thus deposited as dry-deposition.

ILL-EFFECTS

- Acid rain is harmful for agriculture, trees and plants as it dissolves and washes away nutrients needed for their growth.
- It causes respiratory ailments in human beings and animals.
- When acid rain falls and flows as ground water to reach rivers, lakes etc. it affects plants and animal life in aquatic ecosystem.
- It corrodes water pipes resulting in the leaching of heavy metals such as iron, lead and copper into the drinking water.
- Acid rain damages buildings and other structures made of stone or metal. The Taj Mahal in India has been affected by acid rain

How We Can Help To Reduce The Formation Of Acid Rain.

- reducing the emission of sulphur dioxide and nitrogen dioxide in the atmosphere.
- use less vehicles driven by fossil fuels; use less sulphur content fossil fuels for power plants and industries.
- use natural gas which is a better fuel than coal or use coal with less sulphur content.
- Catalytic converters must be used in cars to reduce the effect of exhaust fumes on the atmosphere.
- We can also reduce the acidity of the soil by adding powdered limestone to neutralise the acidity of the soil.

Taj Mahal and Acid Rain

- ❖ The air around the city of Agra, where the Taj Mahal is located, contains fairly high levels of sulphur and nitrogen oxides.
- ❖ It is mainly due to a large number of industries and power plants around the area
- ❖ Use of poor quality of coal, kerosene and firewood as fuel for domestic purposes add up to this problem.
- ❖ The resulting acid rain reacts with marble, CaCO₃ of Taj Mahal causing damage to this wonderful monument
- ❖ As a result, the monument is being slowly disfigured and the marble is getting discoloured and lusterless

- ❖ The Government of India announced an action plan in early 1995 to prevent the disfiguring of this historical monument
- ❖ Mathura refinery has already taken suitable measures to check the emission of toxic gases.
- ❖ This plan aims at clearing the air in the 'Taj Trapezium'— an area that includes the towns of Agra, Firozabad, Mathura and Bharatpur
- ❖ Under this plan more than 2000 polluting industries lying inside the trapezium would switch over to the use of natural gas or liquefied petroleum gas instead of coal or oil.
- ❖ A new natural gas pipeline would bring more than half a million cubic metres of natural gas a day to this area.
- ❖ People living in the city will also be encouraged to use liquefied petroleum gas in place of coal, kerosene or firewood.
- ❖ Vehicles plying on highways in the vicinity of Taj would be encouraged to use low sulphur content diesel.

2. Particulate Pollutants

- Particulates pollutants are the minute solid particles or liquid droplets in air.
- These are present in vehicle emissions, smoke particles from fires, dust particles and ash from industries. Particulates in the atmosphere may be viable or non-viable.
 - *Smoke particulates consist of solid or mixture of solid and liquid particles formed during combustion of organic matter.* Examples are cigarette smoke, smoke from burning of fossil fuel, garbage and dry leaves, oil smoke etc.
 - *Dust is composed of fine solid particles (over $1\mu\text{m}$ in diameter), produced during crushing, grinding and attribution of solid materials. **Sand from sand blasting, saw dust from wood works, pulverized coal, cement and fly ash from factories, dust storms etc.,** are some typical examples of this type of particulate emission.*
 - *Mists are produced by particles of spray liquids and by condensation of va herbicides and insecticides that miss their targets and travel through air and form mists.*
 - Fumes are generally obtained by the condensation of vapours during sublimation, distillation, boiling and several other chemical reactions. Generally, organic solvents, metals and metallic oxides form fume particles.
- The effect of particulate pollutants are largely dependent on the particle size. Airborne particles such as dust, fumes, mist etc., are dangerous for human health
- Particulate pollutants bigger than 5 microns are likely to lodge in the nasal passage, whereas particles of about 10 micron enter into lungs easily.
- Lead used to be a major air pollutant emitted by vehicles. Leaded petrol used to be the primary source of air-borne lead emission in Indian cities.

- This problem has now been overcome by using unleaded petrol in most of the cities in India.
- Lead interferes with the development and maturation of red blood cells.

Smog

The word smog is derived from smoke and fog. This is the most common example of air pollution that occurs in many cities throughout the world

There are two types of smog:

- (a) **Classical smog** occurs in cool humid climate. It is a mixture of smoke, fog and sulphur dioxide. Chemically it is a reducing mixture and so it is also called as reducing smog.
- (b) **Photochemical smog** occurs in warm, dry and sunny climate. The main components of the photochemical smog result from **the action of sunlight on unsaturated hydrocarbons and nitrogen oxides produced by automobiles and factories.**

Photochemical smog has high concentration of oxidising agents and is, therefore, called as oxidising smog.

Formation of photochemical smog

- When fossil fuels are burnt, a variety of pollutants are emitted into the earth's atmosphere. Examples are sulphuric acid mist and troposphere.
- Two of the pollutants that are emitted are hydrocarbons (unburnt fuels) and nitric oxide (NO).
- When these pollutants build up to sufficiently high levels, a chain reaction occurs from their interaction with sunlight in which NO is converted into nitrogen dioxide (NO₂).
- This NO₂ in turn absorbs energy from sunlight and breaks up into nitric oxide and free oxygen atom
- (i) Oxygen atoms are very reactive and combine with the O₂ in air to produce ozone.
- (ii) The ozone formed in the above reaction (ii) reacts rapidly with the NO(g) formed in the reaction to regenerate NO₂. NO₂ is a brown gas and at sufficiently high levels can contribute to haze.
- (iii) Ozone is a toxic gas and both NO₂ and O₃ are strong oxidising agents and can react with the unburnt hydrocarbons in the polluted air to produce chemicals such as **formaldehyde, acrolein and peroxyacetyl nitrate (PAN).**

Effects of photochemical smog

- The common components of photochemical smog are **ozone, nitric oxide, acrolein, formaldehyde and peroxyacetyl nitrate (PAN).**
- Photochemical smog causes serious health problems.
- Both ozone and PAN act as powerful eye irritants
- Ozone and nitric oxide irritate the nose and throat and their high concentration causes headache, chest pain, dryness of the throat, cough and difficulty in breathing
- Photochemical smog leads to cracking of rubber and extensive damage to plant life. It also causes corrosion of metals, stones, building materials, rubber and painted surfaces

How can photochemical smog be controlled?

- control the primary precursors of photochemical smog, such as NO₂ and hydrocarbons, the secondary precursors such as ozone and PAN, the photochemical smog will automatically be reduced.

- Usually **catalytic converters** are used in the **automobiles**, which prevent the release of nitrogen oxide and hydrocarbons to the atmosphere.
- Certain plants e.g., **Pinus, Juniparus, Quercus, Pyrus and Vitis** can metabolise nitrogen oxide and therefore, their plantation could help in this matter.

Stratospheric Pollution

Formation and Breakdown of Ozone

- The upper stratosphere consists of considerable amount of ozone (O₃), which protects us from the harmful ultraviolet (UV) radiations (λ 255 nm) coming from the sun.
- These radiations cause skin cancer (melanoma) in humans. Therefore, it is important to maintain the ozone shield.
- Ozone in the stratosphere is a product of UV radiations acting on dioxygen (O₂) molecules.
- The UV radiations split apart molecular oxygen into free oxygen (O) atoms. These oxygen atoms combine with the molecular oxygen to form ozone. $O_2(g) \xrightarrow{UV} O(g) + O(g)$ $O(g) + O_2(g) \rightarrow O_3(g)$ Ozone is thermodynamically unstable and decomposes to molecular oxygen.
- there have been reports of the depletion of this protective ozone layer because of the presence of certain chemicals in the stratosphere.
- The main reason of ozone layer depletion is believed to be the release of chlorofluorocarbon compounds (CFCs), also known as freons.
- These compounds are nonreactive, non flammable, non toxic organic molecules and therefore used in refrigerators, air conditioners in the production of plastic foam and by the electronic industry for cleaning computer parts etc.
- Once CFCs are released in the atmosphere, they mix with the normal atmospheric gases and eventually reach the stratosphere.
- In stratosphere, they get broken down by powerful UV radiations, releasing chlorine free radical.
- The chlorine radicals are continuously regenerated and cause the breakdown of ozone.
- Thus, CFCs are transporting agents for continuously generating chlorine radicals into the stratosphere and damaging the ozone layer.

The Ozone Hole

- In 1980s atmospheric scientists working in Antarctica reported about depletion of ozone layer commonly known as **ozone hole over the South Pole**.
- In summer season, **nitrogen dioxide and methane react with chlorine monoxide) and chlorine atoms forming chlorine sinks**, preventing much ozone depletion, whereas in winter, special type of clouds called **polar stratospheric clouds** are formed over Antarctica.
- These polar stratospheric clouds provide surface on which chlorine nitrate formed gets hydrolysed to form hypochlorous acid .It also reacts with hydrogen chloride to give molecular chlorine

- When sunlight returns to the Antarctica in the spring, the sun's warmth breaks up the clouds and HOCl and Cl₂ are photolysed by sunlight,
- The chlorine radicals thus formed, initiate the chain reaction for **ozone depletion**

Effects of Depletion of the Ozone Layer

- With the depletion of ozone layer, more UV radiation filters into troposphere.
- UV radiations lead to ageing of skin, cataract, sunburn, skin cancer, killing of many phytoplanktons, damage to fish productivity etc
- plant proteins get easily affected by UV radiations which leads to the harmful mutation of cells.
- It also increases evaporation of surface water through the stomata of the leaves and decreases the moisture content of the soil
- Increase in UV radiations damage paints and fibres, causing them to fade faster.

WATER POLLUTION

- Easily identified source or place of pollution is called as point source. e.g., municipal and industrial discharge pipes where pollutants enter the water-source

Non point sources of pollution

- are those where a source of pollution cannot be easily identified, e.g., agricultural run off (from farm, animals and crop-lands), acid rain, storm-water drainage (from streets, parking lots and lawns),

Causes of Water Pollution

(i) Pathogens:

- The most serious water pollutants are the disease causing agents called pathogens.
- Pathogens include bacteria and other organisms that enter water from domestic sewage and animal excreta.

Human excreta

- contain bacteria such as **Escherichia coli and Streptococcus faecalis which cause gastrointestinal diseases.**

(ii) Organic wastes:

- The other major water pollutant is organic matter such as leaves, grass, trash etc.
- They pollute water as a consequence of run off.
- Excessive phytoplankton growth within water is also a cause of water pollution. These wastes are biodegradable
- The large population of bacteria decomposes organic matter present in water. They consume oxygen dissolved in water. That is why even a moderate amount of organic matter when decomposes in water can deplete the water of its dissolved oxygen
- dissolved oxygen in water is very important for aquatic life . If the concentration of dissolved oxygen of water is below 6 ppm, the growth of fish gets inhibited.

- The dissolved oxygen is also used by microorganisms to oxidise organic matter.
- If too much of organic matter is added to water, all the available oxygen is used up. This causes oxygen dependent aquatic life to die.
- Thus, anaerobic bacteria (which do not require oxygen) begin to break down the organic waste and produce chemicals that have a foul smell and are harmful to human health.
- Aerobic (oxygen requiring) bacteria degrade these organic wastes and keep the water depleted in dissolved oxygen.
- Thus, the amount of oxygen required by bacteria to break down the organic matter present in a certain volume of a sample of water, is called **Biochemical Oxygen Demand (BOD)**
- **The amount of BOD in the water is a measure of the amount of organic material in the water, in terms of how much oxygen will be required to break it down biologically.**

(iii) **Chemical Pollutants:**

- water is an excellent solvent, water soluble inorganic chemicals that include **heavy metals such as cadmium, mercury, nickel** etc constitute an important class of pollutants.
- All these metals are dangerous to humans because our body cannot excrete them. Over the time, it crosses the tolerance limit
- These metals then can damage **kidneys, central nervous system, liver** etc.
- Acids (like sulphuric acid) from mine drainage and **raw salt** used to melt snow and ice in the colder climates (**sodium and calcium chloride**) are **water soluble chemical pollutants**.
- Petroleum products pollute many sources of water e.g., major oil spills in oceans.
- Other organic substances with serious impacts are the pesticides that drift down from sprays or runoff from lands.
- Various industrial chemicals like **polychlorinated biphenyls, (PCBs)** which are used as cleansing solvent, detergents and fertilizers add to the list of water pollutants. PCBs are suspected to be carcinogenic.
- Fertilizers contain phosphates as additives.
- The addition of phosphates in water enhances algae growth. Such profuse growth of algae, covers the water surface and reduces the oxygen concentration in water.
- This leads to anaerobic conditions, commonly with accumulation of abnoxious decay and animal death.
- Thus, bloom-infested water inhibits the growth of other living organisms in the water body.

- This process in which nutrient enriched water bodies support a dense plant population, which kills animal life by depriving it of oxygen and results in subsequent loss of biodiversity is known as Eutrophication.

International Standards for Drinking Water

The International Standards for drinking water are given below and they must be followed.

- **Fluoride:** For drinking purposes, water should be tested for fluoride ion concentration.
- Its deficiency in drinking water is harmful to man and causes diseases such as tooth decay etc.
- Soluble fluoride is often added to drinking water to bring its concentration **upto 1 ppm**. The F⁻ ions make the **enamel on teeth much harder**
- However, F⁻ ion concentration **above 2 ppm causes brown mottling of teeth.**
- At the same time, **excess fluoride (over 10 ppm) causes harmful effect to bones and teeth**, as reported from some parts of Rajasthan.

Lead: Drinking water gets contaminated with lead when lead pipes are used for transportation of water. The prescribed upper limit concentration of lead in drinking water is about 50 ppb. Lead can **damage kidney, liver, reproductive system etc.**

Sulphate: Excessive sulphate (>500 ppm) in drinking water causes **laxative effect**, otherwise at moderate levels it is harmless.

Nitrate: The maximum limit of nitrate in drinking water is 50 ppm. Excess nitrate in drinking water can cause **disease such as methemoglobinemia ('blue baby' syndrome)**

SOIL POLLUTION

insecticides, pesticides and herbicides cause soil pollution.

Pesticides

- Prior to World War II, many naturally occurring chemicals such as nicotine (by planting tobacco plants in the crop field), were used as pest controlling substance for major crops in agricultural practices
- .During World War II, DDT was found to be of great use in the control of malaria and other insect-borne diseases.
- Therefore, after the war, DDT was put to use in agriculture to control the damages caused by insects, rodents, weeds and various crop diseases.
- However, due to adverse effects, its use has been banned in India.
- Pesticides are basically synthetic toxic chemicals with ecological repercussions. The repeated use of the same or similar pesticides give rise to pests that are resistant to that group of pesticides thus making the pesticides ineffective.
- Therefore, as insect resistance of DDT increased, other organic toxins such as Aldrin and Dieldrin were introduced in the market by pesticide industry.

- Most of the organic toxins are water insoluble and nonbiodegradable.
- These high persistent toxins are, therefore, transferred from lower trophic level to higher trophic level through food chain
- Over the time, the concentration of toxins in higher animals reach a level which causes serious metabolic and physiological disorders
- a new series of less persistent or more bio-degradable products called organo-phosphates and carbamates have been introduced in the market
- But these chemicals are severe nerve toxins and hence more harmful to humans.
- As a result, there are reports of some pesticides related deaths of agricultural field workers.
- Insects have become resistant to these insecticides also
- The insecticide industry is engaged in developing new groups of insecticides.
- These days, the pesticide industry has shift sodium chlorate (NaClO_3), sodium arsenite (Na_3AsO_3) and many others.
- During the first half of the last century, the shift from mechanical to chemical weed control had provided the industry with flourishing economic market.
- But one must remember that these are also not environment friendly. Most herbicides are toxic to mammals but are not as persistent as organo-chlorides
- These chemicals decompose in a few months.
- Like organo-chlorides, these too become concentrated in the food web.
- Some herbicides cause birth defects.
- Studies show that cornfields sprayed with herbicides are more prone to insect attack and plant disease than fields that are weeded manually.

INDUSTRIAL WASTE

- Industrial solid wastes are also sorted out as biodegradable and non-degradable wastes. Biodegradable wastes are generated by cotton mills, food processing units, paper mills, and textile factories.
- Non-biodegradable wastes are generated by thermal power plants which produce fly ash; integrated iron and steel plants which produce blast furnace slag and steel melting slag
- Industries manufacturing aluminium, zinc and copper produce mud and tailings. Fertilizer industries produce gypsum.
- Hazardous wastes such as inflammables, composite explosives or highly reactive substances are produced by industries dealing in metals, chemicals, drugs, pharmaceuticals, dyes, pesticides, rubber goods etc.

- The disposal of non-degradable industrial solid wastes, if not done by a proper and suitable method, may cause serious threat to the environment.
- New innovations have led to different uses of waste material
- Nowadays, fly ash and slag from the steel industry are utilised by the cement industry.
- Large quantities of toxic wastes are usually destroyed by controlled incineration, whereas small quantities are burnt along with factory garbage in open bins.
- Moreover, solid wastes if not managed effectively, affect the components of the environment

STRATEGIES TO CONTROL ENVIRONMENTAL POLLUTION

Waste Management

- ✓ Solid waste is not the only waste in household garbage box. there are medical, agricultural, industrial and mining wastes.
- ✓ improper disposal of wastes is one of the major causes of environmental degradation. Therefore, the management of wastes is of utmost importance.

Collection and Disposal

- ✓ Domestic wastes are collected in small bins, which are then transferred to community bins by private or municipal workers.
- ✓ From these community bins, these are collected and carried to the disposable site. At the site, garbage is sorted out and separated into biodegradable and non-biodegradable materials.
- ✓ Non-biodegradable materials such as plastic, glass, metal scraps etc. are sent for recycling.
- ✓ Biodegradable wastes are deposited in land fills and are converted into compost.
- ✓ The waste if not collected in garbage bins, finds its way into the sewers.
- ✓ Non-biodegradable wastes like polythene bag, metal scraps, etc. choke the sewers and cause inconvenience.
- ✓ Polythene bags, if swallowed by cattle can cost their lives also.
- ✓ The poor management causes health problems leading to epidemics due to contamination of ground water.
- ✓ It is specially hazardous for those who are in direct contact with the waste such as rag pickers and workers involved in waste disposal, as they are the ones who handle waste materials mostly without protective device such as gloves or water proof boots and gas masks.

GREEN CHEMISTRY

- ✓ **Green chemistry** is a way of thinking and is about utilising the existing knowledge and principles of chemistry and other sciences to reduce the adverse impact on environment.
- ✓ Utilisation of existing knowledge base for reducing the chemical hazards along with the developmental activities is the foundation of green chemistry.
- ✓ **Green chemistry** , is a cost effective approach which involves reduction in material, energy consumption and waste generation

Green Chemistry in day-to-day Life

(i) Dry Cleaning of Clothes

- ❖ Tetra chloroethene ($\text{Cl}_2\text{C}=\text{CCl}_2$) was earlier used as solvent for dry cleaning.
- ❖ The compound contaminates the ground water and is also a suspected carcinogen
- ❖ The process using this compound is now being replaced by a process, where liquefied carbondioxide, with a suitable detergent is used.
- ❖ Replacement of halogenated solvent by liquid CO_2 will result in less harm to ground water.

(ii) Bleaching of Paper

- ❖ Chlorine gas was used earlier for bleaching paper. These days, hydrogen peroxide (H_2O_2) with suitable catalyst, which promotes the bleaching action of hydrogen peroxide, is used.
- ❖ hydrogen peroxide (H_2O_2) is used for the purpose of bleaching clothes in the process of laundry, which gives better results and makes use of lesser amount of water

Strategies for controlling environmental pollution can be:

- (i) **waste management** i.e., reduction of the waste and proper disposal, also recycling of materials and energy,
- (ii) adopting methods in day-to-day life, which results in the reduction of environmental pollution.

Thanks for Reading
Don't forget to Share it with the Needy Aspirant.